

EC441 Midterm Two Spring 2022

Name 1: _____ Name 2: _____ Version 406

This is a closed book exam. You may use one 8.5x11 piece of paper, typed or written on both sides, that can be read normally (no micro printing). Calculators are allowed. The time for the exam is 90 minutes.

TCP (20 points)

- (a) If you connect a raspberry-PI to a fast desktop computer using a 1 Gb/s Ethernet link, and attempt to send a large file from the desktop to the PI, will TCP congestion control or TCP flow control be the more important mechanism to make this work properly? Explain.
- (b) Suppose the round trip time between a web client and a web server is 5 ms. What is the usual delay before a small web page can be displayed by the web client, starting from the time the URL is entered into the browser by the user? Assume that TCP is used and that this is the first request from this client to this server.
- (c) Suppose you are sending packets with 1000 bytes of application layer data from A to B. A receives a packet with the ACK number of 12345 and a sequence (SEQ) number of 54321. What would the next packet that A receives be ? (i.e. what would the ACK and SEQ numbers be?) If there are several likely options, explain.

SOLUTION

- (a) There is no need for congestion control as there is only one sender. Flow control is likely to be necessary because the PI as the receiver probably cant keep up with the flow of information coming into its TCP socket receive buffer.
- (b) The TCP connection must be established first. This requires one RTT, and the file request and response also requires a RTT. Hence the usual delay is about 10 ms.
- (c) The ACK number would be 13345, which is 1000 bytes higher than the previous one. The SEQ number is likely to be 54321 again, since B is not sending data back to A during this process.

Networks and Hosts (30 points)

- (a) Which of the following are valid IPv4 networks? If it is a valid network, what is the CIDR representation (i.e. a.b.c.d/x)?

Net Address	Net Mask	Valid?
10.0.0.0	255.0.0.0	10.0.0.0/8
192.168.0.0	255.255.255.0	192.168.0.0/24
128.197.128.0	255.255.192.0	128.197.128.0/18
128.197.128.9	255.255.192.0	invalid
128.197.128.40	255.255.255.248	128.197.128.40/29

- (b) Someone tells you about their IPv4 network over the phone, but you only hear this:

187.45.____6.0/19

You are missing zero, one, or two digits where the blank is: the third number could be 6, 86, 186, 986, etc.

What is the correct value for the third number?

- (c) You are an ISP and have been assigned the network 191.45.12.0/22. You wish to break this into /24 and /25 networks. Provide an example of a decomposition of the original network that uses all the addresses and includes both /24 and /25 networks.

SOLUTION

- (a) See table above.
- (b) For a /19 network, the first three bits of the third byte can be on or off. So the third number must be a multiple of 32 (there are 8 of these). The options are 0,32,64,96,128,160,192,224.

The third number must be 96.

- (c) When going from /22 to /24, the last two bits of the third number can be set. This adds 0,1,2, or 3 to the network. In this design, let's use 3 of the '/24' and the remaining one will be split into two /25 networks. These /25 have the first bit of the fourth byte either on (128) or off (0).

```
[IPv4Network('191.45.12.0/24'),  
 IPv4Network('191.45.13.0/24'),  
 IPv4Network('191.45.14.0/24'),  
 IPv4Network('191.45.15.0/25'),  
 IPv4Network('191.45.15.128/25')]
```

Error Control (25 points)

- (a) Design an error control code with 2 data bits that can be used for single bit error detection. Use the fewest number of redundancy bits as possible. Your answer should be in the form of a set of n-bit codewords, where your task is to determine both n and the codewords.
- (b) Design an error control code with 2 data bits that can be used for single bit error correction. Use the fewest number of redundancy bits as possible. Your answer should be in the form of a set of n-bit codewords, where your task is to determine both n and the codewords.
- (c) What is the minimum distance achieved by the IP checksum coding scheme? Could IP do error correction using its current checksum structure?

SOLUTION

- (a) We need to add a parity bit. The code will be

```
000  
011  
101  
110
```

- (b) Can this be done with 2 redundancy bits? Start with 0000. The next word must be 1110. There is no third word that is 3 away from both.

With 3 redundancy bits, it is possible, as follows:

```
00000  
01110  
10101  
11011
```

- (c) The IP checksum adds 16-bit values of the packet. Changing one bit in the packet changes just one bit of the checksum, so the minimum distance is 2. Hence error correction is not possible.

Routing and AS (25 points)

- (a) Consider a network with 24-bit addresses for the hosts. The maximum size of a subnet is 256 hosts. The routing table entries are of the form a.b.c/x (dotted decimal with CIDR notation). This means that all the hosts that are inside a subnet are represented by a single routing table entry.

You would like the routing tables of all routers (inter-AS routers, intra-AS routers, gateway routers) to be no larger than 500 entries.

How can this be accomplished? How would you break the network into autonomous systems? What should the size of the ASes be?

- (b) Find the shortest path between node A and node g in the following network. Describe the path like this: ABcZg. What is the cost of this path?
- (c) If the link cost from g to h changed from 3 to 1, would the path from the previous question change? If so, what would the new path be?

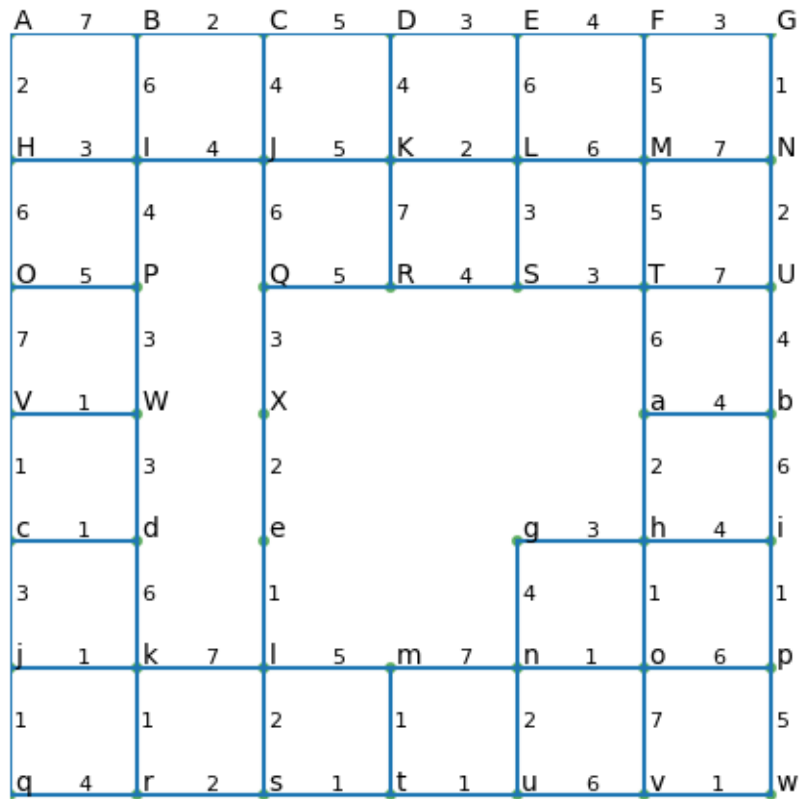


Figure 1: Network with costs indicated for Problem 4

SOLUTION

- (a) To minimize the number of networks and hence the size of the routing tables, let's assume the subnets are all 256 hosts and so we have $2^{24}/256 = 2^{16}$ networks. The simplest first attempt is to make 256 ASes each of which have 256 networks. But in this scenario, some routers (the gateway routers) need 512 entries (one for each AS and one for all the local networks in the AS). So we need to break the system into a third level with hierarchical intra-AS routing. If we have 2^8 networks (256), then break each AS into $2^4 = 16$ area routers each of which have $2^4 = 16$ local networks, then the maximum number of entries in my routing table will be $256 + 16 + 16 < 500$
- (b) Using Dijkstra (see Figure 2), we find this path: AHIPWVcjkrtung at a cost of 29.
- (c) Yes. See Figure 2. The new path would be AHIPWVcjkrtunohg for a cost of 28.

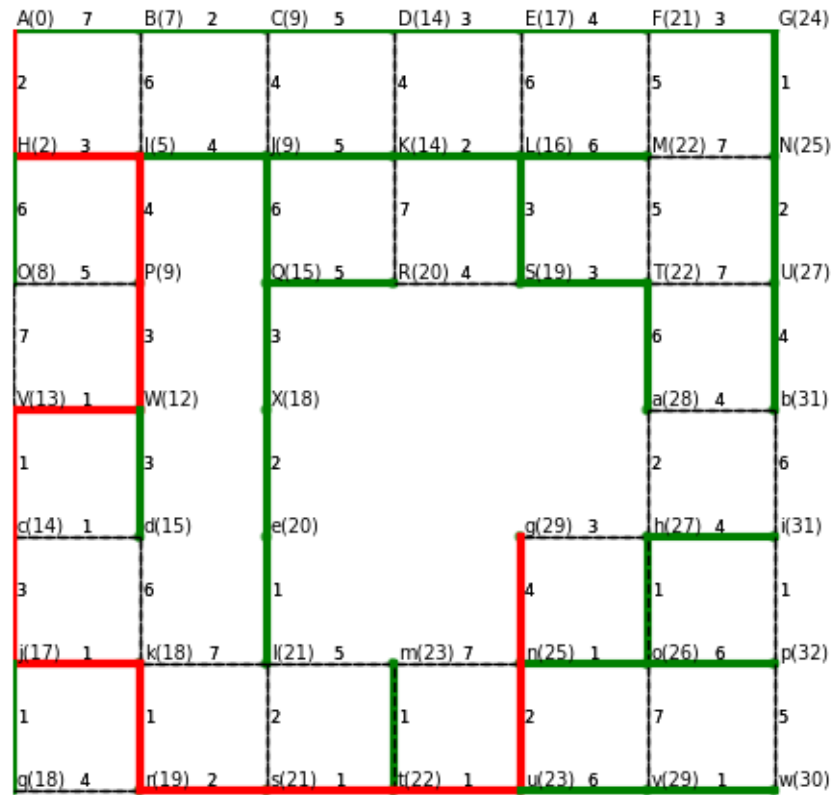


Figure 2: Solution for path from A to g